

FTP & TFTP

- Topics covered:
 - The Purpose of FTP/TFTP
 - FTP/TFTP Functions & Differences
 - IOS File System
 - Using FTP/TFTP in IOS

The Purpose of FTP/TFTP

Use of FTP and TFTP

FTP and TFTP are industry-standard client-server protocols used to transfer files across a network.

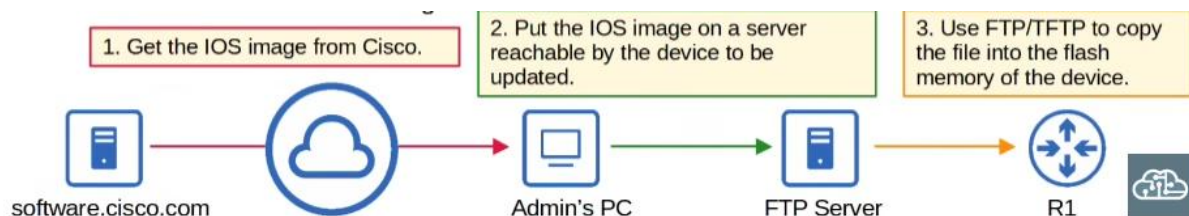
A client can:

- Copy files **from** an FTP/TFTP server.
- Copy files **to** an FTP/TFTP server.

A common use in networking is **upgrading Cisco IOS**:

1. Download the IOS image from Cisco.
2. Place the image on a server reachable by the Cisco device.
3. Copy the image into the device's flash memory using FTP or TFTP.
4. Reboot the device using the new IOS image.

IOS Upgrade Process



TFTP (Trivial File Transfer Protocol)

What is Trivial File Transfer Protocol?

TFTP was first standardized in **1981**.

It is called "**Trivial**" because it provides only basic file transfer functionality. Unlike FTP, it only allows a client to:

- Copy a file from a server.
- Copy a file to a server.

It does not provide any additional file management capabilities.

TFTP is **not** a replacement for FTP. Instead, it is intended for situations where a lightweight and simple protocol is preferred.

Security Characteristics

TFTP provides no security features.

- No usernames or passwords.
- No authentication.
- No encryption.
- All data is transmitted as plain text.

Because of this, it is best suited for controlled environments where small files need to be transferred quickly.

Port Number

TFTP servers listen on: UDP port 69

Clients send their initial request to destination UDP port **69**.

TFTP Reliability

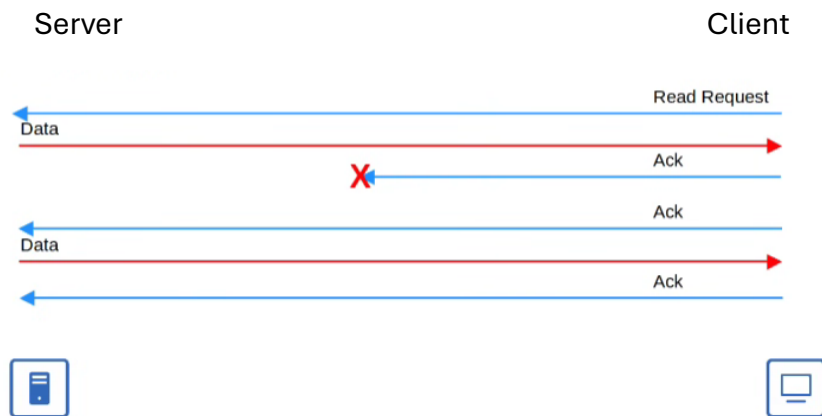
Although UDP itself is connectionless and unreliable, TFTP implements its own reliability.

Every data message is acknowledged.

- If the client receives data, it sends an acknowledgment (Ack).
- If the server receives data, it sends an acknowledgment.
- Timers detect missing messages.
- If an expected message does not arrive, the previous message is retransmitted.

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Lock-Step Communication



TFTP uses lock-step communication.

The client and server alternate sending messages:

- One device sends a message.
- It waits for the reply.
- Only after receiving the reply does it send the next message.

Retransmissions occur whenever necessary.

TFTP File Transfer Phases

A TFTP transfer consists of three phases.

1. Connection
 - The client sends a request.
 - The server replies, initializing the connection.
2. Data Transfer
 - Data messages and acknowledgments are exchanged.
3. Connection Termination
 - After the final data message, a final acknowledgment is sent.
 - The connection is terminated.

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Transfer Process



Although TFTP does not use TCP, it includes a basic mechanism for managing connections within the protocol.

TFTP Transfer Identifier (TID)

This topic is beyond the CCNA scope but explains part of TFTP's operation.

Initially:

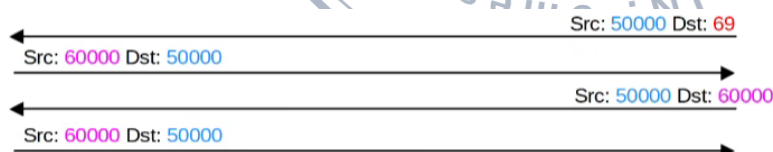
- Client source port = random ephemeral port (called the Transfer Identifier **(TID)**).
- Destination port = **UDP 69**.

When the server replies:

- The server also selects a **random TID** as its source port.
- It does **not** use port 69.

All subsequent communication uses the client and server TIDs.

Port Usage



Port 69 is used only for the client's initial message.

FTP (File Transfer Protocol)

What is File Transfer Protocol?

FTP was first standardized in **1971**.

Unlike TFTP:

- Uses TCP.
- Uses authentication.
- Provides many file management functions in addition to file transfers.

FTP uses:

- **TCP port 21** — Control connection
- **TCP port 20** — Data connection

Like TFTP, FTP does not encrypt data.

Both usernames/passwords and transferred data are sent in plain text.

For greater security:

- FTPS (FTP Secure) adds SSL/TLS.
- SFTP (SSH File Transfer Protocol) is a different protocol with a similar name.

FTP Capabilities

FTP provides more functionality than TFTP.

Clients can:

1. Transfer files.
2. Navigate directories.
3. Add directories.
4. Remove directories.
5. List files.
6. Send various FTP commands.

Unlike FTP, TFTP clients can only:

1. Upload files.
2. Download files.

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FTP Connections

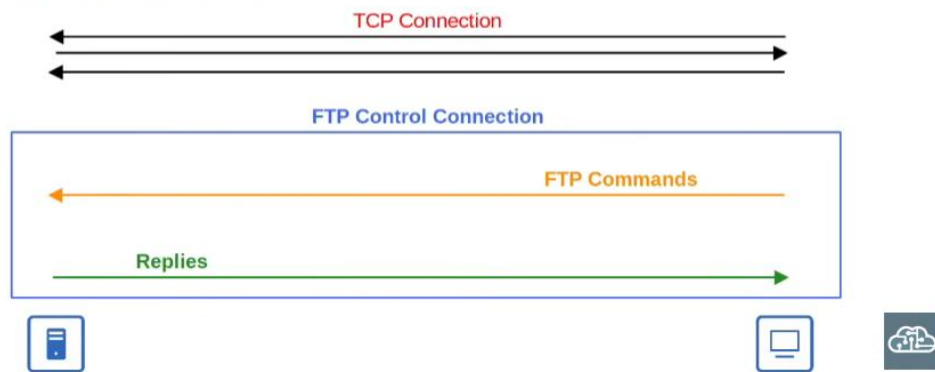
FTP uses two separate TCP connections.

1. **Control connection (TCP 21)**
 - Used for FTP commands and server replies.
2. **Data connection (TCP 20)**
 - Created only when transferring data.
 - Established and terminated as needed.

FTP Control Connection

FTP uses two types of connections:

- An **FTP control** connection (**TCP 21**) is established and used to send FTP commands and replies.
- When files or data are to be transferred, separate **FTP data** (**TCP 20**) connections are established and terminated as needed.



FTP Active Mode

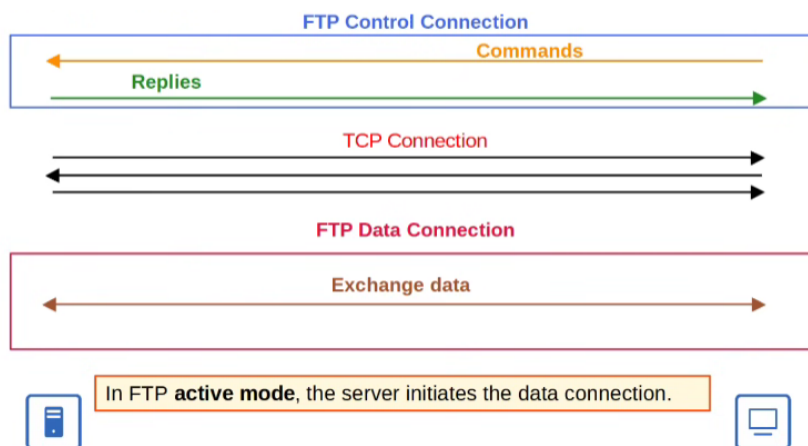
Active mode is the default method.

The process is:

- The FTP control connection already exists.
- The **server initiates** the TCP data connection.
- Data is exchanged.
- The control connection remains active during the transfer.

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Active Mode



Key point: In FTP active mode, the server initiates the data connection.

FTP Passive Mode

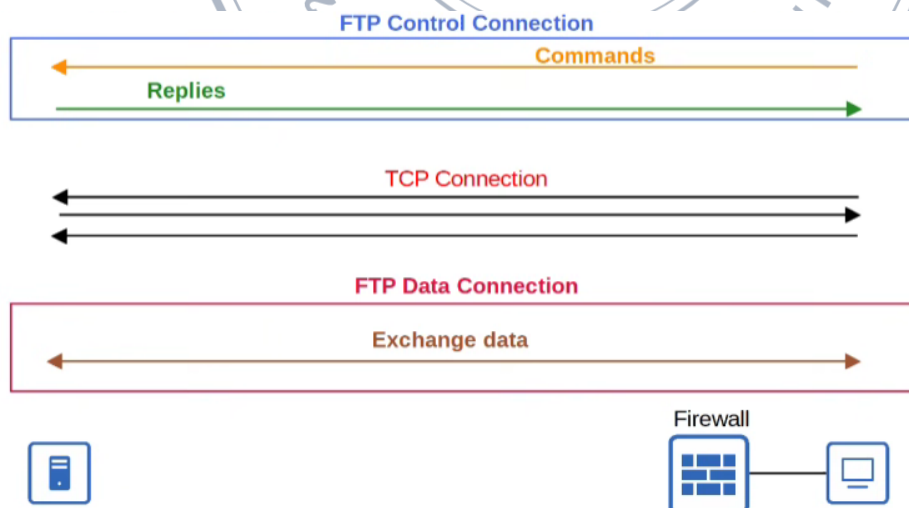
Passive mode is commonly used when the client is behind a firewall.

Many firewalls do not allow outside devices to initiate incoming connections.

Therefore:

- The **client** initiates the data connection.
- The firewall allows the server's replies.
- The server no longer initiates the connection.

Passive Mode



Key point: In FTP passive mode, the client initiates the TCP data connection because firewalls often block connections initiated from outside devices.

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FTP vs TFTP Comparison

FTP

Uses TCP

TCP port 21 (control), TCP port 20 (data)

Connection-oriented

Supports many FTP commands

Can navigate directories and manage files

Uses usernames and passwords

More complex

TFTP

Uses UDP

UDP port 69

UDP with basic connection handling inside the protocol

Only uploads and downloads files

Cannot list, delete, or manage files

No authentication

Simpler

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IOS File System

Cisco IOS File Systems

A **file system** controls how data is stored and retrieved. Cisco IOS devices contain multiple file systems, which can be viewed using:

R1# show file systems

	Size(b)	Free(b)	Type	Flags	Prefixes
* 2142715904	1994403840	-	disk	rw	flash0: flash:#
-	-	-	disk	rw	flash1:
966656	962560	-	disk	rw	flash2:#
-	-	-	disk	rw	flash3:
-	-	-	opaque	rw	archive:
-	-	-	opaque	rw	system:
262144	256791	-	nvram	rw	nvram:
-	-	-	opaque	rw	tmpsys:
-	-	-	network	rw	snmp:
-	-	-	opaque	rw	null:
-	-	-	network	rw	tftp:
-	-	-	opaque	ro	xmodem:
-	-	-	opaque	ro	ymodem:
-	-	-	opaque	wo	syslog:
-	-	-	network	rw	rcp:
-	-	-	network	rw	pram:
-	-	-	network	rw	ftp:

[output omitted]

disk: Storage devices such as flash memory.

opaque: Used for internal functions

nvram: Internal NVRAM. The startup-config file is stored here.

network: Represents external file systems, for example external FTP/TFTP servers.

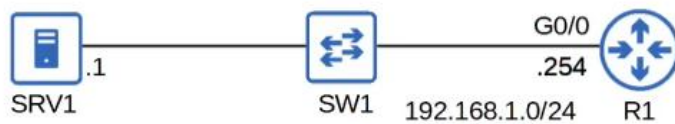
Common Cisco IOS File System Types

Type	Purpose
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disk	Storage devices such as flash memory, where the IOS image is usually stored. During boot, the IOS image is copied from flash into RAM.
opaque	Used for internal logical functions rather than physical storage devices.
nvram	Non-Volatile RAM. Unlike normal RAM, it retains data after power loss. The startup-config is stored here.
network	Represents external file systems such as FTP and TFTP servers.

Using FTP/TFTP in IOS

Demonstration Topology



1. SRV1 stores a newer Cisco IOS image.
2. The server provides both TFTP and FTP services.
3. R1 downloads the IOS image into its flash memory.

Viewing the Current IOS Version

```
R1#show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE
(fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team
[output omitted]
```

Notes

1. Displays the IOS image currently running.
2. Example image: **C2900-UNIVERSALK9-M**
3. K9 indicates support for cryptographic features such as **SSH**.
4. Current version: **15.1(4)M4**

Only the beginning of the command output is shown because the full output is much longer.

Viewing Flash Memory

```
R1#show flash
System flash directory:
File Length Name/status
3 33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
2 28282 sigdef-category.xml
1 227537 sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

This command displays the contents of flash memory, including the IOS image currently stored on the router.

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Step 3 — Configure the Boot Image

Enter global configuration mode:

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#boot system flash:c2900-universalk9-mz.SPA.155-3.M4a.bin
R1(config)#exit
R1#write memory
Building configuration...
[OK]
R1#reload
Proceed with reload? [confirm]
```

Important: If boot system is not configured, the router boots using the first IOS image it finds in flash, Saving the configuration before reloading is required; otherwise, the boot system command does not take effect.

Step 4 — Verify the Upgrade

After the router restarts:

R1# show version

The output now shows the newer IOS version (**15.5**) instead of **15.1**.

```
R1#show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.5(3)M4a, RELEASE SOFTWARE(fc1)
[output omitted]
```

Step 5 — Delete the Old IOS Image

Use:

R1# delete flash:c2900-universalk9-mz.SPA.151-4.M4.bin

The router asks for confirmation before deleting the file.

```
R1#delete flash:c2900-universalk9-mz.SPA.151-4.M4.bin
Delete filename [c2900-universalk9-mz.SPA.151-4.M4.bin]?
Delete flash:/c2900-universalk9-mz.SPA.151-4.M4.bin? [confirm]
```

Verify the result:

R1# show flash

```
R1#show flash

System flash directory:
File Length Name/status
  4 33591768 c2900-universalk9-mz.SPA.155-3.M4a.bin
  2  28282 sigdef-category.xml
  1 227537 sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

The older IOS image no longer appears in flash.

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Upgrading IOS Using TFTP

The upgrade procedure is almost identical. Only the file transfer step changes.

Before copying files, configure the FTP credentials that the router will use.

```
R1(config)# ip ftp username cisco
```

```
R1(config)# ip ftp password cisco
```

The same username and password must also exist on the FTP server.

```
R1(config)#ip ftp username cisco
R1(config)#ip ftp password cisco
R1(config)#exit
```

Configure the FTP username/password that the device will use when connecting to an FTP server.

Copy the file: R1# copy ftp: flash:

The router then prompts for:

- FTP server IP address
- Source filename
- Destination filename

```
R1#copy ftp: flash:
Address or name of remote host []? 192.168.1.1
Source filename []? c2900-universalk9-mz.SPA.155-3.M4a.bin
Destination filename [c2900-universalk9-mz.SPA.155-3.M4a.bin]?

Accessing ftp://192.168.1.1/c2900-universalk9-mz.SPA.155-3.M4a.bin...
Loading c2900-universalk9-mz.SPA.155-3.M4a.bin from
192.168.1.1: !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[output omitted]
```

After the file is copied, the remaining steps are exactly the same:

1. Configure **boot system**
2. Save the configuration
3. Reload the router

Commands Summary

```
R1# show file systems
R1# show version
R1# show flash
R1# copy source destination
R1(config)# boot system filepath
R1(config)# ip ftp username username
R1(config)# ip ftp password password
```